FEDERAL STANDARD STOCK CATALOG

Section IV

(Part 5)

FEDERAL SPECIFICATION

FOR

LACES, SHOE; COTTON

This specification was approved on the above date by the Director of Procurement for the use of all departments and establishments of the Government, and shall become effective not later than January 15, 1947 for use in procurement initiated on or after that date. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE SPECIFICATION

A–1. The following Federal Specification, of the issue in effect on date of invitation for bids, shall form a part of this specification:

CCC–T–191—Textiles; General Specifications, Test Methods.

B. TYPES, CLASSES, LENGTHS, COLORS, AND GRADE

B–1. *Types.*—Cotton shoe laces in accordance with this specification shall be furnished in the following types, as specified in the invitation for bids:

Type I. —Mercerized.
Type II. —Soft.
Type III.—Glazed.
Type IV.—Waxed.

B–2. *Classes.*—Cotton shoe laces shall be furnished in the classes indicated in table I for each type, as specified in the invitation for bids. The classes are based on differences in yarn size, number of carriers, and number of picks per inch.

B–3. *Lengths.*—Cotton shoe laces shall be furnished in the lengths indicated in table I, as specified in the invitation for bids.

B–4. *Colors.*—Cotton shoe laces shall be furnished in the following colors, as specified in the invitation for bids:

Black	Dark brown	Olive drab
White	Medium brown	Khaki
	Russet brown	

B–5. *Grade.*—Cotton shoe laces shall be of the grade known commercially as "firsts."

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—Cotton shoe laces shall be made from thoroughly cleaned cotton free from waste.

C–2. *Workmanship.*—Workmanship shall be first class. The laces shall be free from obvious imperfections and defects or blemishes.

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D. GENERAL REQUIREMENTS

D–1. *Construction.*—Cotton shoe laces shall be tubular-braided without a core and pressed flat.

D–2. *Tips.*—Each end of the shoe laces shall be completely encircled by a firmly attached tip which shall match the color of the braid and which shall be smooth, without sharp edges or projections. The tips shall be made of fiber, slow-burning plastic, or nonferrous metal.

D–2a. *Dimensions.*—The length of the tips shall be 9/16 inch minimum, 13/16 inch maximum. If of metal, the thickness shall be a minimum of 0.010 inch.

D–2b. *Gripping strength.*—Tips shall not be dislodged from their position on the laces, when subjected to the test described in paragraph F–3b.

D–3. *Tolerances.*—

D–3a. *Length.*—A tolerance of minus ¼ inch or plus ½ inch shall be allowed in the length of the laces.

D–3b. *Width.*—A tolerance of plus or minus 1/64 inch shall be allowed in the finished width of the laces.

D–4. *Color fastness.*—Cotton shoe laces shall show at least "fair" color fastness when tested for fastness to light, water, and rubbing (crocking) in accordance with Federal Specification CCC–T–191.

E. DETAIL REQUIREMENTS

E–1. *Type I, mercerized.*—The laces shall be braided of mercerized cotton yarn; the finished lace shall have a smooth and silky luster.

E–2. *Type II, soft.*—The laces shall be braided of soft cotton yarn.

E–3. *Type III, glazed.*—The laces shall be braided of soft cotton yarn which has been glazed; the finished braid shall have a smooth, glossy finish.

E–4. *Type IV, waxed.*—The laces shall be braided of soft cotton yarn; the finished braid shall have a wax finish and shall be soft and pliable.

E–4a. Waxing shall be accomplished by immersion in a molten wax solution. (See par. E–4b.) The wax shall permeate to the inside of the braid and shall be evenly distributed throughout the braid.

E–4b. *Wax solution.*—The solution shall consist of 9.25 pounds minimum, 11.70 pounds maximum, of copper naphthenate containing 10 percent of copper computed as metal, added to each 100 pounds of a wax mixture compounded as follows:

	Percent:
Micro-crystalline wax (yellow, 165° F. m.p.)	5.25
Beeswax (yellow)	5.25
Carnauba wax (N.C. ref.)	36.85
Paraffin wax (135° F. m.p.)	36.85
Ceresin wax (175° F. m.p.)	15.80
	100.00

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E–4c. After passing through the waxing process, the braid shall increase in weight not less than 40 percent nor more than 55 percent, based on the finished weight of the braid, and shall contain not less than 0.3 percent copper computed as metal.

E–5. *Physical properties.*—The length, width, construction, and breaking strength shall conform to the requirements in table I.

TABLE I.—*Physical properties.*

Type	Class	Length	Width (finished)	Yarn size and ply (approx.)	Ends per carrier	No. of carriers (min.)	No of picks per inch (min.)	Breaking strength (min.)
		Inches	*Inches*					*Pounds*
I-Mercerized	A	18 21 24 27 28½	7/32	54/2	2	44	40	70
	B	24 27 30 33 36 40 54	7/32	50/2	2	44	32	70
	C	24 27 30 33 36 38 40	7/32	40/2	2	64	40	70
	D	28½ 31½	7/32	16/2 to 20/2	2	40	26	90

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TABLE I.—*Physical properties. (Continued)*

Type	Class	Length	Width (finished)	Yarn size and ply (approx.)	Ends per carrier	No. of carriers (min.)	No. of picks per inch (min.)	Breaking strength (min.)
		Inches	*Inches*					*Pounds*
II-Soft	E	30	8/32	20/2	2	32	16	120
	F	60	11/32	20/2	2	44	16	150
III-Glazed	G	27 30 40	8/32	20/2	2	32	16	120
	H	36 40 54 60	11/32	20/2	2	44	16	150
IV-Waxed	I	40 54	7/32	24/2	2	40	22	110
	J	40 52	7/32	26/2	2	44	22	110
	K	40 54	7/32	30/2	2	48	26	110
	L	40	10/32	28/2 to 30/2	2	44	22	125

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Sampling.*—Samples for inspection and test shall be taken at random in an amount equal to 1 percent of each lot, but not less than 6 laces nor more than 20. For the purpose of this specification, a lot shall be considered as consisting of that quantity of laces of identical type, class, length, and color offered for inspection at any one time or being within a single shipment.

F–2. *Inspection.*—Inspection shall be made at point of delivery, unless otherwise specified in the invitation for bids, to determine compliance with this specification with respect to construction, workmanship, and color.

F–3. *Tests.*—

F–3a. Tests for color fastness, breaking strength, width, and length, shall be conducted in accordance with Federal Specification CCC–T–191, with the following modifications:

F–3a(1). *Breaking strength.*—For the breaking strength test, the two pairs of jaws shall have smooth grips and shall be set 6 inches apart at the start of the test.

F–3a(2). *Measurement of length.*—The length of the tips shall be included in the length of the laces, and the laces shall be measured from end to end when suspended vertically under a tension of 3 ounces.

F–3b. *Test for tips.*—

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F–3b(1). A rigid metal plate having a narrow slot is required. The slot shall be not less than 4 inches long, ¼ inch wide at one end and 1/32 inch wide at the other end. The slot shall be smoothly finished. The rigid plate does not necessarily have to be a single piece; for instance, the slot may be formed by a clamped or bolted assembly of two metal bars or plates properly spaced apart.

F–3b(2). The plate shall be supported in a horizontal position. The lace shall be passed upward, tip first, through the wide end of the slot until the tip clears the plate, and then moved horizontally toward the narrow end of the slot until the end of the tip rests on top of the metal. To the hanging end of the lace, about 10 inches from the tip, a 20-pound weight shall be gently attached.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging.*—Unless otherwise specified, commercial packages are acceptable.

G–2. *Packing.*—Unless otherwise specified, the subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to the point of delivery called for in the contract or purchase order.

G–3. *Marking.*—

G–3a. *Issue packages.*—Issue packages shall be marked with the name of the material, the type, class, length, and color, and the name or trade-mark of the contractor.

G–3b. *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name of the material, the type, class, length, and color, and the quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS

H–1. The following departmental specifications, of the issue in effect on date of invitation for bids, and special requirements, shall form a part of this specification, and shall be applicable to purchases made under this specification by the respective departments:

H–2. *Army.*—

H–2a. *Applicable specifications.*—

Federal Specification:

UU–P–268—Paper; Kraft, Wrapping.

Joint Army-Navy Specifications:

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

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U. S. Army Specification:

100–2—Standard Specification for Marking Shipments by Contractors.

Copies of the specifications listed above may be obtained without cost upon application to the procuring agency.

H–2b. *Packaging.*—

H–2b(1). *Laces, except mukluk boot laces (type II).*—One hundred and forty-four laces shall be tied together in one lot, and five lots shall be tied in one bundle.

H–2b(2). *Mukluk boot laces (type II).*—Thirty-six of the shorter laces and thirty-six of the longer laces shall be tied together in one lot, and ten lots shall be tied in one bundle.

H–2b(3). Each bundle specified above shall be wrapped in grade B kraft paper, 60-pound basis weight (24 x 36 — 500) conforming to Federal Specification UU–P–268.

H–2b(4). *Labeling.*—Each bundle shall be legibly marked with the following information:

Stock number
Nomenclature
Length of lace
Quantity
Name of manufacturer
Contract number
Specification number

H–2c. *Packing.*—

H–2c(1). *For domestic shipment.*—Unless otherwise specified in the contract or order, laces shall be packed in style RSC, regular slotted containers, type CF or SF, grade 5 or 6, conforming to Joint Army-Navy Specification JAN–P–108, except that strapping will not be required.

H–2c(2). *For overseas shipment.*—Laces for overseas shipment shall be packed in style 4 nailed wood boxes conforming to and strapped in accordance with Joint Army-Navy Specification JAN–P–106. Boxes shall be lined with a waterproof barrier material conforming to and sealed in accordance with Joint Army-Navy Specification JAN–P–125, type L–2 or M.

H–2d. *Marking for shipment.*—Shipping containers shall be marked in accordance with U. S. Army Specification 100–2.

H–2e. The following types and classes of laces covered by this specification are listed in the Quartermaster Supply Catalog as follows:

Type I, class A.—Laces, Shoe, Low-Quarter, Tan, 28½-inch—Stock No. 72–L–80.

Type I, class B.—Laces, Shoe, Low-Quarter, Tan, 30-inch—Stock No. 72–L–81.

Type I, class C.—Laces, Shoe, Field, Women's, 38-inch—Stock No. 72–L–74–30.

Type I, class C.—Laces, Shoe, Service, Women's, Low, 27-inch—Stock No. 72–L–87.

Type I, class C.—Laces, Nurses', White, 27-inch—Stock No. 72–L–81–60.

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Type II, class E.—Laces, Boot, Mukluk, 30-inch—Stock No. 72–L–46–20.

Type II, class F.—Laces, Boot, Mukluk, 60-inch—Stock No. 72–L–46–25.

Type IV, class I.—Laces, Shoe, Service, Cotton, 40-inch—Stock No. 72–L–85.

Type IV, class J.—Laces, Boot, Ski-Mountain, 52-inch—Stock No. 72–L–47–44.

H–3. *Marine Corps.*—The color of the laces shall match the Marine Corps standard sample.

H–4. *Navy.*—

H–4a. *Applicable specification and publication.*—General Specifications for Inspection of Material and the Navy Shipment Marking Handbook, copies of which may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J.

H–4b. *Marking.*—In addition to any special marking required by the contract or order, marking of interior packages and shipping containers shall conform to the requirements of the latest issue of the Navy Shipment Marking Handbook.

H–4c. *Breaking strength.*—Method described in Federal Specification CCC–T–191 applies except that the distance between smooth jaws shall be 3 inches at the start of test.

I. NOTES

I–1. *Intended uses.*—

I–1a. Type I laces are intended for use with dress shoes; type II laces are for mukluks; type III laces are for service or work shoes; type IV laces are for shoes used chiefly outdoors where extra resistance to weather is required.

I–1b. Twenty-four-inch to thirty-inch length laces are for use chiefly with oxfords (low-quarter shoes); 36-, 38-, and 40-inch length laces are for high shoes such as service, garrison, or combat shoes; 52-, 54-, and 60-inch length laces are for 8- and 10-inch height boots such as ski and jungle boots.

I–2. Purchasing officers should exercise any desired options offered herein and should specify the type, class, length, and color required. (See pars. B–1, B–2, B–3, B–4, G–1, G–2, G–3b, H–2c.)

I–3. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be required prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

I–4. Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the specifications, or which are commercially available, but are intended to

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cover the types, etc., which are suitable for Federal Government requirements.

I–5. An Index of Federal Specifications may be purchased as noted in the paragraph below, price to be obtained from the Superintendent of Documents.

I–6. Copies of this specification and of CCC–T–191 and Supplement thereto may be obtained upon application, accompanied by postal note, money order, coupon, or cash, to Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price of this specification and CCC–T–191, 5 cents each; Supplement to CCC–T–191, 10 cents.

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